

# Alexander Reinthal

📍 Gothenburg, Sweden    ✉ email@reinthal.me    ☎ 070-918 68 50    🌐 reinthal.me    📄 alexander-reinthal  
📱 reinthal

---

Senior cyber security practitioner and data engineer switching careers to mitigate loss of control risks through AI control research.

## Research Projects

---

### Out of Control Arena

Sept 2026 – present

Developed a control arena setting to compare AI's capability in cyber defense compared to offense.

- Ported BashArena to a VM setting (Proxmox)
- Extended monitors to perform agentic forensics on victim machines

### Aletheia's Quest: Lie detection Hackathon

July 2026 – Aug 2026

Developed a hybrid lie detector with EleutherAI that generalized over 27 different datasets, 3 different model families and many different types of lies.

- Contributed to new datasets and validating data generation, labeling of data and writing the blog post.
- Awarded 2nd place on the validation set.

### Can lying probes flag a Chinese model's deceptive answers?

Jan 2026 – June 2026

Found that some Chinese models think they are honest when answering politically censored questions.

- Extension of ARENA 7.0 weekend hackathon, 1st place. Supported by Bluedot Impact rapid grant.

### About Emergent Misalignment

June 2026 – present

Found that emergent misalignment (EM) rate is negatively correlated with the capability benchmarks GSM8K (Pearson  $-0.98$ ) and MMLU (Pearson  $-0.94$ ).

- Extension of ARENA 7.0 capstone, 3rd place.

### Changing the prompt for instrumental convergence evals

May 2026 – June 2026

Removing pressure phrases increases misalignment rate for capability prompts ([He et al. 2025](#)) but decreases it for propensity prompts ([Hopman et al. 2026](#)) for Gemini 3.1 Pro Preview.

### Detecting Piecewise Cyber Espionage in Model APIs

Nov 2025 – Nov 2025

Proof of concept for detecting malicious activity across multiple agent sessions.

- Research continued through SPAR by David Williams-King (ERA) and Linh Le (Mila). Apart Research sprint, 4th place of 641 submissions.

## Experience

---

### Career Transition Grantee

May 2026 – present

Independent

- Supported by Coefficient Giving's Career Transition grant for reducing global catastrophic risks. Currently researching lie detection with a broader interest in AI control and scalable oversight agendas.

### Research fellow

Sept 2026 – Dec 2026

SPAR

- Researching Lie detection under Walter Laurito and Kieron Kretschmar (Cadenza Labs).

## Alum

Sept 2026 – Sept 2026

*AI Security Bootcamp*

- Studied an intense curriculum across AI Safety, AI security and Technical AI governance.

## Research mentor

Mar 2026 – present

*Bluedot Impact*

- Facilitator and research mentor for the Technical AI Safety course and project.

## Alum

Jan 2026 – Feb 2026

*ARENA 7.0*

- Five-week intensive curriculum on transformer internals, mechanistic interpretability, RLHF, evals, JAX, and the Inspect framework.

## Forward-deployed Engineer / DevSecOps / Tech Lead / Data Scientist

Gothenburg, Sweden

*Knowit Solutions Cocreate*

Feb 2022 – present

- Lead secure deployment of data platforms across multiple clients, some under protective security contracts ([Säkerhetsskyddslagen 2018:585](#)).
- Pioneered LLM evals at the company. Implemented an OAuth-authenticated MCP gateway to secure agentic workflows.

## Cyber Security Analyst

Gothenburg, Sweden

*NTT Security*

Apr 2019 – Jan 2022

- Wrote calibrated cyber incidents reports based on available evidence. Deployed an ML classifier that reduced analysis time of malicious domains from 2 hours to 10 minutes per analyst per shift.

## Python Software Engineer

Gothenburg, Sweden

*Ericsson*

June 2017 – Oct 2018

- First job where I led a team building a log-parsing and analytics tool that got funded and grew into a dedicated engineering team.

## Publications

---

### Data Modeling for Predicting Software Exploits

Nov 2018

Peer-reviewed paper predicting which CVEs become exploited in the wild. Adopted by Recorded Future's Vulnerability Intelligence product.

Reinthal, A., Filippakis, E., Almgren, M.. [10.1007/978-3-030-03638-6\\_21](#) (Springer LNCS: Nordic Conference on Secure IT Systems)

## Open Source and Personal Projects

---

### [Casually Jailbreaking Gemini 2.5 Flash](#)

2026 – 2026

Jailbroke Gemini 2.5 Flash with a few-shot in-context continuation attack, eliciting harmful recommendations.

### Open Source Contributions

[github.com/reinthal](https://github.com/reinthal)

- [Inspect AI PR #3987](#) Fixed Gemini 3.1 Pro + OpenRouter + Reasoning issue
- [Dagster PR #24188](#) (ML pipeline orchestration) and [DLT Hub PR #594](#) (ingestion reliability)

## Skills

---

**Machine Learning & AI:** PyTorch, JAX (learned during ARENA 7.0), Einops, XGBoost, SciKit Learn, Jupyter; LLM evals & observability (Inspect, LangChain, Langfuse, Arize Phoenix); linear probes, SFT, automated red-teaming

**Programming Languages:** Python, JavaScript, SQL, Rust, C/C++, R, Nix, Terraform

**Platforms & Tools:** Modal, Runpod, Openweights, Weights & Biases, Spark, Docker, Kubernetes, AWS, Azure

**Certifications:** Bluedot Impact — Technical AI Safety (2026-03); AGI Strategy (2025-10)

## Education

---

**Chalmers University of Technology**

*MS in Engineering Physics*

*Gothenburg, Sweden*

*Sept 2016 – June 2018*

- Master thesis got published and put into production at Recorded Future.
- Switched from computer science undergraduate to engineering physics graduate program and performed better.

**Gothenburg University**

*BS in Computer Science*

*Gothenburg, Sweden*

*Sept 2013 – June 2016*

- Thesis on finding common dense subgraphs using linear programming has 4 citations.